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Quiz 4

$$T \cdot \left(\frac{r}{r_1} \right)^n = \left(\frac{r}{r_1} \right)^n T_1$$

$$\left(\frac{P_{AB}}{P_{AB,1}} \right) = \left(\frac{T}{T_1} \right)^{3/2}$$

$$T = T_1 \left(\frac{r}{r_1} \right)^n$$

$$P_{AB} = P_{AB,1} \left(\frac{T}{T_1} \right)^{3/2}$$

$$P_{AB,1} \left(\left(\frac{r}{r_1} \right)^n \right)^{3/2}$$

$$c = T \cdot P / RT$$

$$c_{AB}(r) = \frac{P \sqrt{\left(\frac{r}{r_1} \right)^n} P_{AB,1}}{RT_1}$$

$$N_A r = \frac{N_A r_1 (r_1)^2}{r^2} = \frac{X_A(r) N_A r_1 (r_1)^2}{r^2} - \frac{P \sqrt{\left(\frac{r}{r_1} \right)^n} P_{AB,1} \left(\frac{2}{3} X_A(r) \right)}{RT_1}$$

$$X_A = \frac{\left(-2 \frac{RT_1 N_A r_1}{P P_{AB,1} (n+2)} \frac{r^2}{r} \left(1 - \frac{1}{3} n \right) \right)}{1 - e}$$

$$X_{A1} = 1 - X_{B1} \quad X_{A2} = 1 - X_{B2}$$

$$-\frac{1}{2} \frac{\ln \left(\frac{X_{B1}}{X_{B2}} \right) r_2^{(1+1/2n)}}{\left(r_2^{(1+1/2n)} - r_1^{(1+1/2n)} \right) r_1 RT_1} \frac{P P_{AB,1} (n+2)}{r_1 RT_1}$$

Mass balance?

$$= 4 \frac{\pi P P_{AB,1} \left(1 + \frac{1}{2} n \right) \ln \left(\frac{X_{B2}}{X_{B1}} \right)}{RT_1 \left(r_1^{(1+1/2n)} - r_2^{(1+1/2n)} \right) r_1^{(1/2n)}}$$

BCs?

The Diffusion in an non-isothermal film in which the temp changes shows that as the value of n increases, -diffusion increases so. Does the magnitude.